LLM-Driven Evolutionary Search: Evolving packing programs

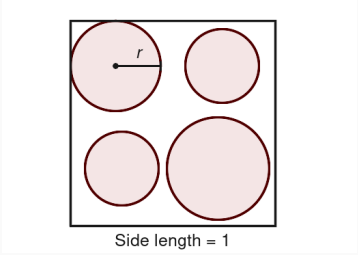
Can we improve the programs that find good circle packings?

ENGR 689 · Chapter 10 · Texas A&M University

Circle packing: the problem

What are we trying to arrange?

Place **26 or 32 circles** inside a square with side length 1. Choose each circle's **center** (its location) and **radius** (half its width). The radii may differ.



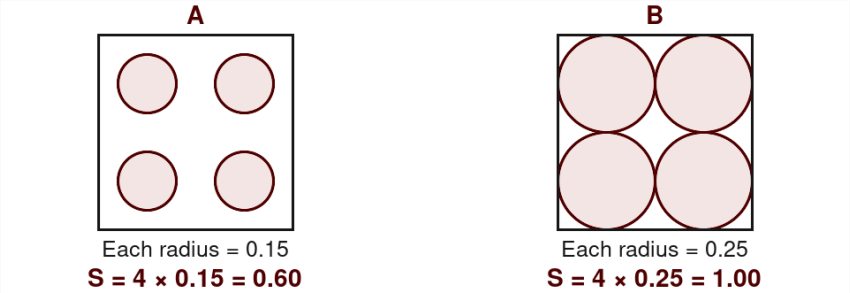
Every circle must stay inside the square. Circles may touch each other and the boundary, but they must **not overlap**.

A valid packing satisfies these rules for every circle. The number of circles stays fixed within each problem.

Circle packing: the goal

Which valid packing is better?

Packing score: $S = r_1 + \cdots + r_N$, the sum of all radii. **Larger is better.**



B has the larger score. Circle positions limit how large the radii can be. We need a strategy to search for better packings.

How a packing program searches

What does a program do to improve one packing?

Example: **Greedy Local Center Search**. Start with circle centers. **Radius allocation** chooses radii for those fixed centers to maximize their sum while satisfying the packing rules.

Propose a random change to one center. The **move scale** controls the typical size of that change.

Recalculate radii and score. The **move acceptance rule** keeps the proposed change only if the score increases. This cycle is one **search step**.

Repeat the steps, then return the best packing found. This loop changes a packing. Later, we will change the program itself.

What is a trial?

What do Trial 1, Trial 2 and Trial 3 mean?

A **trial** is one execution of an unchanged program from given centers. It may contain many search steps. Limits: **80 requested steps**, **8 seconds maximum**.

A **random seed** is a number that initializes the program's random choices. Our defaults are:

Trial	1	2	3
Random seed	3	4	5

Use the same three starting packings and seeds for every program. This shared set of starting packings is the **packing bank**.

Our strategy: evolve packing programs

How can we improve the method that finds the packings?

Run programs to produce packings. **Fitness:** the median (middle) score of three complete, valid trials. Keep useful programs as **parents**.

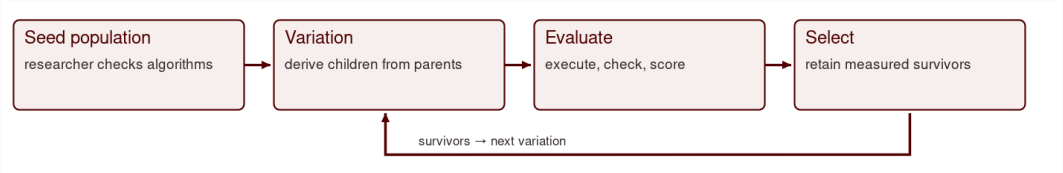


Figure 1: Initialize once; evaluate, select and mutate repeatedly.

Luna, our language model, edits parent code. Test each new program before keeping it.

Five choices define the search

What must we define before running this search?

$(\mathcal{A}, f, \mu, \sigma, P_0)$: allowed programs, measured performance (**fitness**), how programs change (**mutation**), which programs we keep (**selection**), and the starting group (**initial population**).

$$P_{t+1} = \sigma(P_t \cup \{\mu(a) : a \in \sigma(P_t)\}) .$$

Choose parents $\rightarrow$ mutate $\rightarrow$ pool parents and children $\rightarrow$ select survivors.

New programs compete with the parents. A worse new program does not have to replace a useful parent.

Classical variation

How can we change the packing program's code?

Point mutation: edit one constant, such as $0.04 \to 0.02$ for smaller center moves.
Subtree replacement: replace the condition that decides whether to keep a proposed center move.

Crossover: combine code from two parents, such as one program's starting construction with another's move acceptance rule.

These edits change the program, not just one packing. Evaluate the resulting program with the same three-trial protocol before selecting it.

A language model writes the mutation

What can a mutator change when it reads the parent code and its results?

1. Retain the parent's center search and radius allocator.
2. Change center-move size, which centers move, or the move acceptance rule.

3. Luna returns edited **Python source** and a hypothesis.
4. An isolated interpreter executes it; an independent checker evaluates the returned circles.

Luna implements the proposed code change; Python executes it. A reasoned edit becomes evidence only after evaluation.

Context joins the two course spines

Which evidence should the next mutation see?

Parent source **and scores**; successful and failed attempts; task constraints and useful domain knowledge; population similarity and remaining budget.

Evaluation becomes fitness. Context engineering drives mutation. The next request carries the previous round's actual results.

External memory makes the next proposal informed. A plausible explanation does not replace a measured trial.

Elitism protects measured progress

If a mutation is worse, must the current champion disappear?

Pool parents with the new child. Retain the best complete programs under the declared fitness rule.

For a maximized quality, elitism makes the observed best stay level or improve **under comparable evaluation conditions**.

A failed mutation need not erase a useful parent. Preserve the best valid packing separately from program fitness.

Selection pressure and population collapse

Why can keeping only the highest scores make search stagnate?

Top-k: rank everyone and retain the best k. **Tournament:** sample a group and retain its best; smaller tournaments can preserve more alternatives.

Collapse: several survivors become small rewrites of the same search. Current fitness does not reveal every parent's future descendants.

Strong selection exploits evidence but can discard useful routes. Our top-three rule does not guarantee diversity.

Packing score and program fitness

How do we turn packing scores into a program's fitness?

Packing score: $S = \sum_i r_i$, after checking the geometry. A scorer that ignores overlap can reward invalid packings.

Program fitness: run three complete, valid trials with the same starting packings and numerical limits for every program.

$$F = \mathrm{median}(S_1, S_2, S_3).$$

Example scores $1.90,\ 2.10,\ 2.00$ give $F = 2.00$.

The demo maximizes F. Chapter 10 minimizes $f = -F$. Any invalid or incomplete trial makes the program ineligible.

Preserve different search behaviors

How can useful alternatives survive beside the current champion?

Islands: evolve separate populations, with occasional migration. **MAP-Elites:** retain the best program in each behavior bin, such as one-center versus coordinated moves.

Novelty search: favor behaviors unlike those already seen. Different source text alone does not establish different search behavior.

Preserve several ways to search. These are possible extensions; our top-three controller does not implement a diversity archive.

Exploration, exploitation and cost

When should we refine, explore—or stop spending?

Exploit: refine. **Explore:** change methods. **Stop:** respect the cap.

Hypothetical policies, same test conditions:

Policy	Median sum	Numerical seconds
A	2.40	1
B	2.41	5

Illustrative utility $S - 0.01t$ **favors A:** $2.39 > 2.36$**.** Our demo reports costs separately.

Demo 1: follow three mutations

Which part of the parent survives, and does the changed program earn its place?

Evaluate the initial population → inspect parent source and context → send Luna round 1 → read the actual code diff and verified packing.

Repeat for rounds 2 and 3. Inspect inherited packing-bank IDs, complete trial scores, failures and survivor decisions.

Follow the real chain: source → execution → checked geometry → fitness → selection → next context.

Read the first run's evidence

What improved: one packing, the program's measured fitness, or both?

A witness proves one feasible construction. Three complete trials rank a program on one declared bank. A source diff explains what changed; it does not prove an improvement.

Keep construction progress and program comparison separate. Report the actual outcome, calls and numerical effort.

Demo 2: the 32-circle problem

Does the same program-evolution mechanism still work when the number of circles changes?

Use $N = 32$, the second unit-square case in AlphaEvolve Appendix B.12. Keep the objective, checker, trial seeds, source interface and numerical limits declared.

Change the problem instance, then inspect adaptation. A higher raw sum for 32 circles is not evidence of a better algorithm than the 26-circle run.

Demo 2: inspect adaptation and limits

Does Luna change its search after seeing the 32-circle trial evidence?

Evaluate 32-circle parents → send three gated requests → inspect actual source edits, verified packings and the feedback entering each next request.

Keep invalid proposals, incomplete executions and repeated ideas visible. Stop cancels owned work; replay remains explicitly recorded.

The mechanism is testable even when improvement is absent. Evidence decides which proposed programs survive.

Name the framework we observed

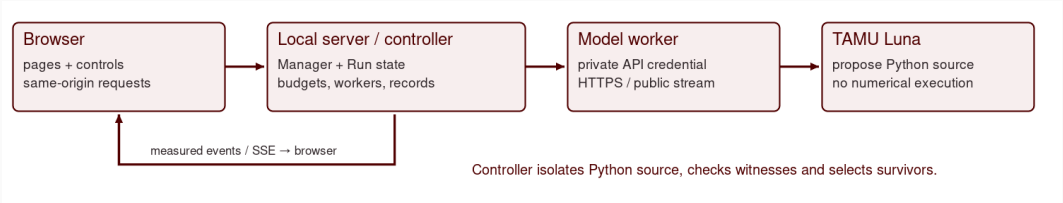
Can you identify every Chapter 10 component in the live record?

$\mathcal{A}$: Python packing programs. f: verified current-bank trial fitness. μ: Luna's code mutation. σ: complete-program selection. P_0: checked initial programs.

Context proposes; external evaluation measures; selection preserves. For your research problem, define these five choices before spending search budget.

Appendix A1: One server coordinates the experiment

Who owns the population, execution and measurements?



The controller owns the loop. Only the model worker contacts TAMU; a separate isolated worker executes generated Python.

Appendix A2: Request, context and source response

What crosses the model API boundary?

Send the task, parent source, current-bank trial results, history and limits. Authorization is a separate HTTPS header.

Return JSON: `round_id`, `parent_ids`, `hypothesis`, `source`. The source defines `solve(initial, seed, steps)`.

A source string is executable code after admission. The hypothesis is an explanation, never the authoritative score.

Appendix A3: Execution, checking and feedback

When can a returned program become a survivor?

Validate source $\rightarrow$ isolated trials $\rightarrow$ verify every circle and pair $\rightarrow$ aggregate three complete scores $\rightarrow$ select within the current packing bank.

Keep exact request/response records, valid witnesses and failures. The next context carries the measured history.

Execution turns a proposal into evidence. Live LLM, numerical-only and recorded replay remain distinct throughout.